It is recommended that you spend about 30 minutes on this question

2 Two students are having a discussion about an experiment in which the air inside a bell jar is gradually removed. The sound of a ringing bell inside the jar is heard to diminish in intensity during this process. One student suggests that the frequency of the sound changes as the pressure changes; the other student thinks that the frequency remains constant as the air is removed.

Design a laboratory experiment to investigate how the frequency of a sound wave depends on the pressure of air. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements that would be taken,

(c) how the frequency of the sound would be measured using a cathode ray oscilloscope,

(d) the control of variables,

(e) any safety precautions that you would take.

© UCLES 2004

https://xtremepape.rs/